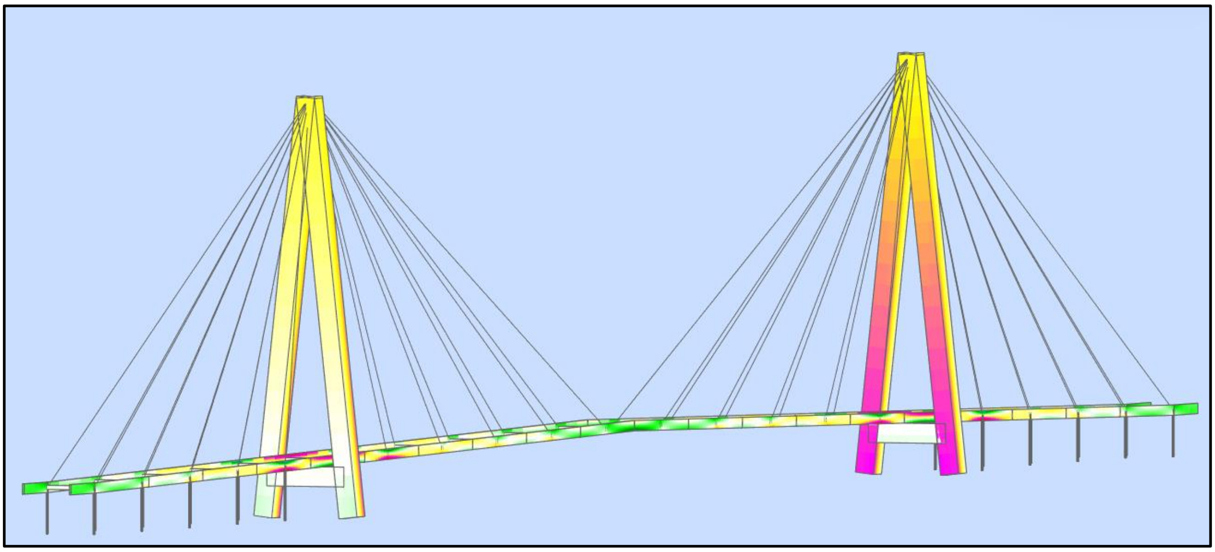
# PLASTIC PERFECTION?

## A BRIEF FINITE ELEMENT ANALYSIS OF A **GFRP** FOOTBRIDGE



f

*AR30400 – Structures 3*

**GROUP 4**


| Name | Signature |
|---|---|
| Jasper Crosland (JDC200) | |
| Francesca Franchi (FF448) | |
| Lewis Acott (LA684) | |
| Daniel Heard (DH915) | |

# 1.0 Introduction and Context

The Aberfeldy Footbridge, spanning the River Tay in Scotland, is the focus for our Finite Element Analysis (FEA) using Autodesk Robot (AR). It was constructed to improve access between two parts of the local golf course and to replace an older structure.

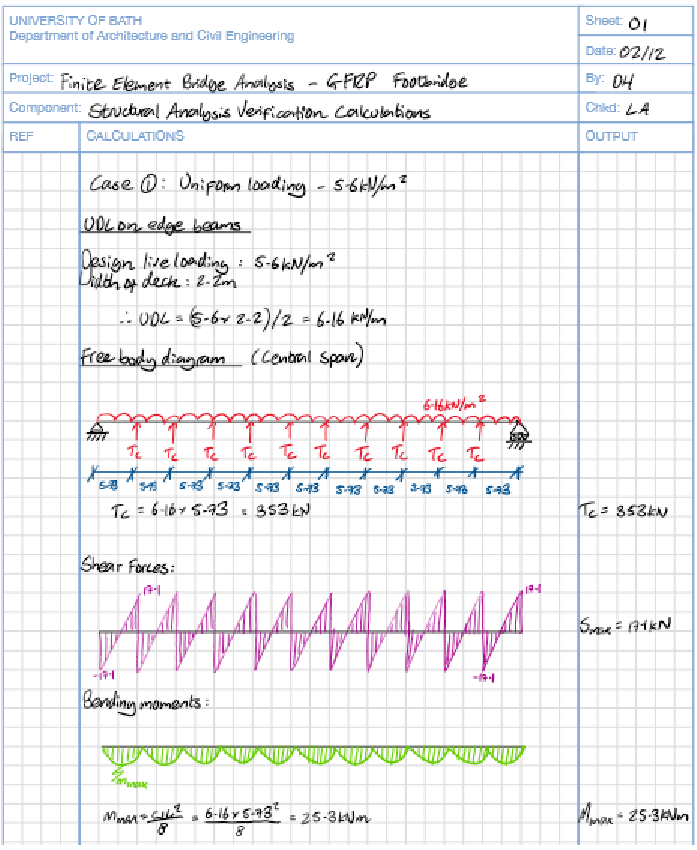
It is a visually striking cable-stayed bridge which is unusual in its construction out of glass fibre-reinforced plastic elements (GFRP) and Kevlar-based cables. When it opened in 1992, it was the longest composite bridge in the world. We chose this design for our study as it was designed by a team of students at Dundee University and has been widely researched, providing lots of technical data to help us model its behaviour more accurately in AR.

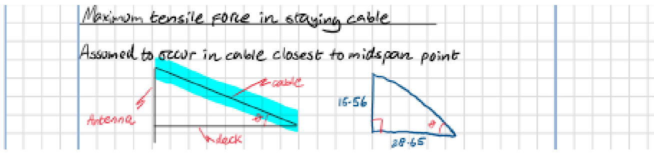
The structural behaviour of the bridge is analysed under various loading scenarios to better understand its performance. In addition to the standard live loading, we gave particular attention to the bridge's response under combined snow and wind loading during a winter storm. Aberfeldy is located on the edge of the Scottish Highlands, an area known for its adverse winter weather conditions making it likely that drifting and high wind loading may happen concurrently.

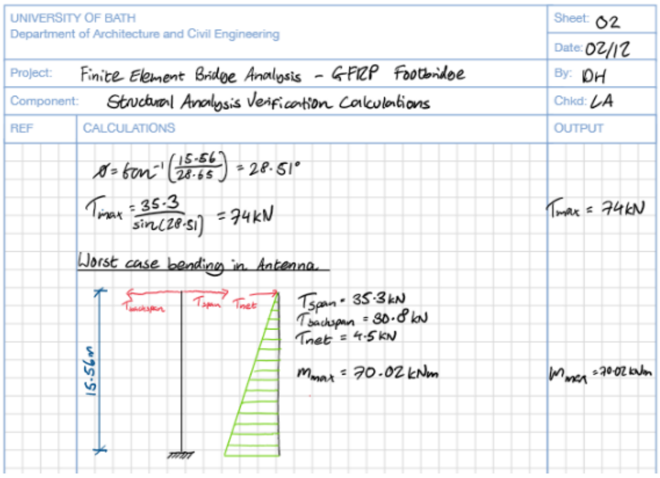
Furthermore, as the bridge forms the link between two sections of the golf course, it may be subjected to concentrated crowd loading during a golf event. Large groups of spectators may gather on smaller sections of the deck to follow the action. Our analysis focuses on how the bridge performs under this asymmetric type of loading.

# 2.0 Analysis - First Principles Structural Mechanics

### 2.1 Everyday Use: Uniform Live Loading – 5.6kN/m$^2$




To reduce the indeterminacy of the structure we have assumed that all the cables are taking the same vertical load. Second-order effects on the supporting antennae were neglected and modelled as pinned supports. This is a conservative assumption as beneficial hogging moments at the supports resulting from the continuation of the beam onto the backspan are not included. The frequent spacing of secondary cross beams (953mm C-C) influenced our decision to model the loading as a UDL rather than point loads. It was also assumed that the deck helps restrain the bending of the antennae caused by the tensile forces in the cables.

### 2.2 Winter Storm: Combined Wind Loading & Snow Drifting

Our method of calculating the bending moments in the edge beams due to snow loading follows the same approach in Section 2.1 (Loads are detailed in Section 4.3). For the wind loading, we assumed the deck is restrained horizontally at the antennae and modelled these as pinned supports.

We assumed that the snow loading would have a negligible effect on the bending of the antennae. Consequently, we calculated the maximum bending moment by modelling it as a cantilever and applying a horizontal UDL. It was assumed that the load transfer between the deck and antenna occurs at deck level. The maximum force in the cable was determined as in section 2.1 with the additional consideration of the horizontal forces.

### 2.3 Golf Tournament: Concentrated Crowd Loading

The same assumptions still apply as set out above. However, the UDL was applied to two deck sections (11.46m) south of the midspan point. We assumed that the load was carried purely by the cables supporting these two deck sections. This is a conservative assumption, as the load is likely to be distributed across the cables in the span.

### 2.4 Predicted Values Table

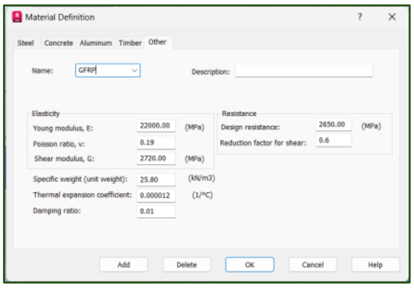
| | Structural Deck Beams | | | | Antennae | | Max Cable Forces kN |
|---|---|---|---|---|---|---|---|
| | Max Bending Moment (Vertical) kNm | Max Stress (MPa) | Max Bending Moment (Horizontal) kNm | Max Stress (MPa) | Max Bending Moment kNm | Max Stress (MPa) | |
| General Live Loading | 25.3 | 62.8 | - | - | 70.08 | 16.7 | 74 |
| Wind & Snow | 4.6 | 11.4 | 928.7* | 6.3x10^3 | 194 | 46.33 | 31 |
| Concentrated Crowd | 25.3 | 62.8 | - | - | 70.08 | 16.7 | 74 |

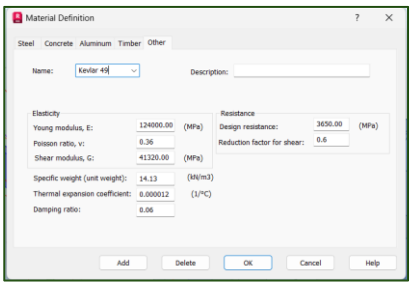
From inspection the assumptions for the horizontal wind loading were significantly over conservative and have resulted in an unrealistically high stress prediction.

## 3.0 Model Setup

### 3.1 Material Generation

Most users of AR take advantage of the built-in materials (Steel, Concrete etc.). However, the software supports the creation of custom user-generated materials for the analysis. The large amount of technical data available for the GFRP components used in the deck, beams and pylons, as well as for the Parafil stay cables with a Kevlar-49 core, made this a feasible option for our study. This data was sourced from many research papers published on the footbridge since its construction. The material properties we used for the bridge are as follows:




### 3.2 Structure Type Selection

A 3D frame analysis was deemed the most appropriate method for modelling the footbridge. While a 2D model would have been easier to construct and more computationally efficient, it would have introduced significant limitations, reducing its ability to accurately predict structural behaviour. A 2D model would not be able to represent the inclined stay cables or the antennae. It would also not effectively model out-of-plane effects such as wind loading.

Additionally, the use of symmetry was considered to cut the amount of modelling required. The center point of the span seemed like an ideal choice, due to the identical structural geometry on either side. However, when we decided to model asymmetrical concentrated crowd loading, it disrupted the loading symmetry, making it impractical to use it in this context.

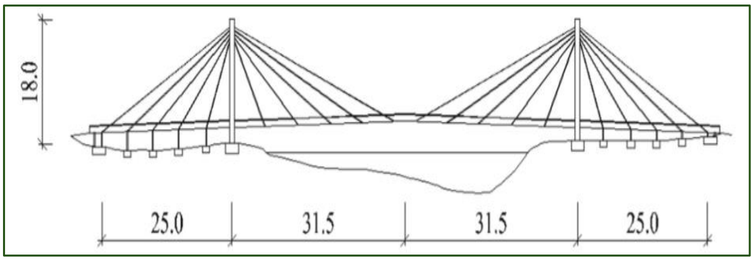
### 3.3 Building the Model

#### 3.3.1 Main Assumptions

Building a finite element model that perfectly represents the real-life behaviour of a complex structure is not practical. Consequently, as engineers, we must use our judgment to make appropriate assumptions that best align the model with the actual structure. For this analysis, we assumed that the GFRP remains within elastic conditions under our loading cases, meaning no permanent deformation or yielding occurs. Architectural details that do not directly affect structural performance have been ignored.

The deck supporting structure, edge beams & primary/secondary cross beams, were modelled but the deck panels themselves were omitted. Instead, the deck's self-weight and any applied loading were transferred directly onto the edge beams. The deck panels, made of timber boards, were not considered to be structural elements and including them would unnecessarily complicate the model making the results unclear.

#### 3.3.2 Model dimensions




Once the nodes were in the model space, we input the three types of deck-supporting beams. All the beams are made of GFRP and are formed out of smaller repeating components. After this, the antennae were added, and the cables and their anchors were connected between the edge beams and the tops of the antennae.

The bases of the antennae have been modelled as fixed supports, while the anchor bases were modelled as pinned. At the southern end of the span, the supports have been modelled as roller supports and at the northern end of the span as pinned.

The two runs of deck edge beams are symmetrical, with a vertical rise at midspan of 1m. The YZ section of the edge beam is 80x400mm, with a thickness of 5mm. The deck spans onto 2200mm long primary beams, which are perpendicular to the edge beams. These primary beams have a 160x160mm section and a thickness of 5mm. They connect to the edge beams at intervals of 5m outside of the antenna supports, and 5.73m in the central span. These connections coincide with the cable anchorage locations.

The antennae have sections of 800x800mm and are 18.11m in length. Their bases are fixed along the ground plane and are braced by 3.62m long 800x800m sections at a height of 2.4m, forming an A-frame structure. In the model, the A-frames are 63m apart.

The cables are anchored with 80mm diameter (5mm thick) aluminium circular hollow sections, spanning from the cable anchorage locations on the deck, down to ground level where they are pinned. These support the back spans, with heights ranging from 2m at each end, and progressively increasing to match the slope of the deck.

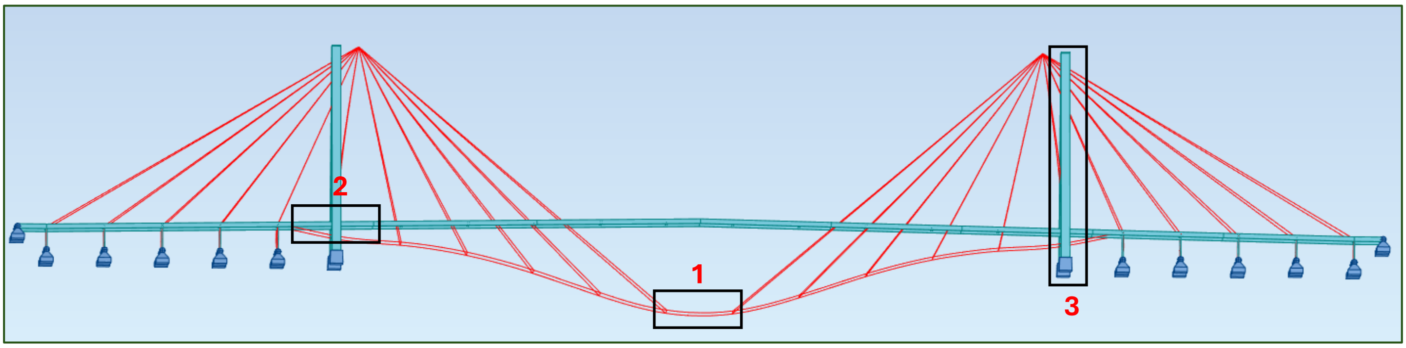
#### 3.3.3 Challenges & Learning Points

When we ran our model for the first time, we encountered the following errors: "Separate Structural System Warning" and multiples of "No convergence to a non-linear solution". This was initially frustrating, as there were no obvious issues with the model. However, we established that the cables were not correctly connected to the main beams or the antennae. Instead of being attached to the common nodes, we attached them to the projected face of the elements. Consequently, they were operating as independent systems and not transferring load between themselves correctly. Additionally, the second issue was caused by a default setting within AR. When cables were introduced to the model, they were automatically given a 100kN tensile pre-stress. Once we removed this, the model converged to a solution which was in line with the expected behaviour from our analysis.

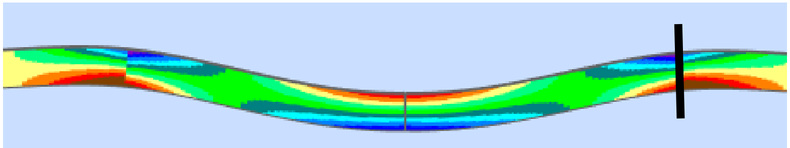
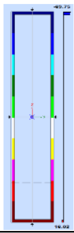
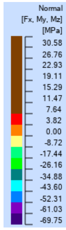
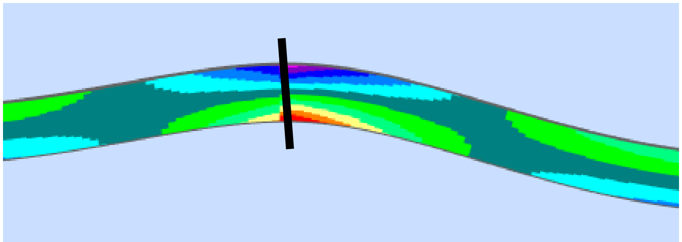
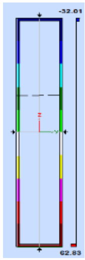
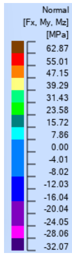
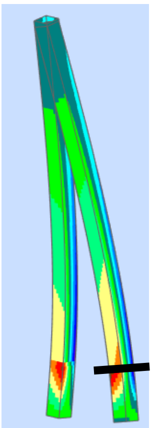
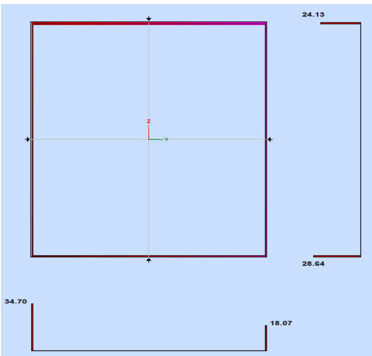
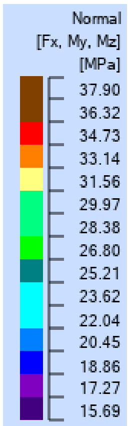
# 4.0 Finite Element Analysis

## 4.1 Normal Design Live Loading

### 4.1.1 Deflection Based Location of Interest Identification



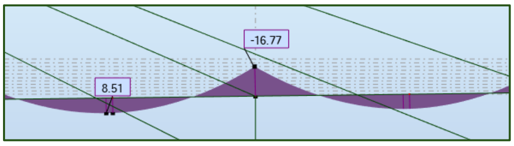

### 4.1.2 Maximum Stress Locations

| | Elevation | YZ Section | Forces Key |
|---|---|---|---|
| 1 |  |  |  |
| 2 |  |  |  |
| 3 |  |  |  |



Our initial analysis maps the locations of high stresses well. The maximum stress of 62.9 MPa in the edge beams from our FEA maps very closely with our predicted value of 62.8 MPa. However, this value is potentially misrepresentative as in our calculations we assume that there are no hogging moments which are present in these results. Additionally, the value of 37.9 in the antenna is much higher than our assumption of 16.7 this is likely caused by our failure to map the localised stresses caused by the edge beams interacting with the antenna. Our hand calculation methodology did not allow for the redistribution of moments which may be the reason for some of the discrepancies. The combined (Sagging & Hogging) bending moment on the edge beams sums to 25.25kNm which is very similar to our prediction of 25.3kNm. Additionally, the FEA accurately shows the maximum tensile force in the staying cables as shown below.

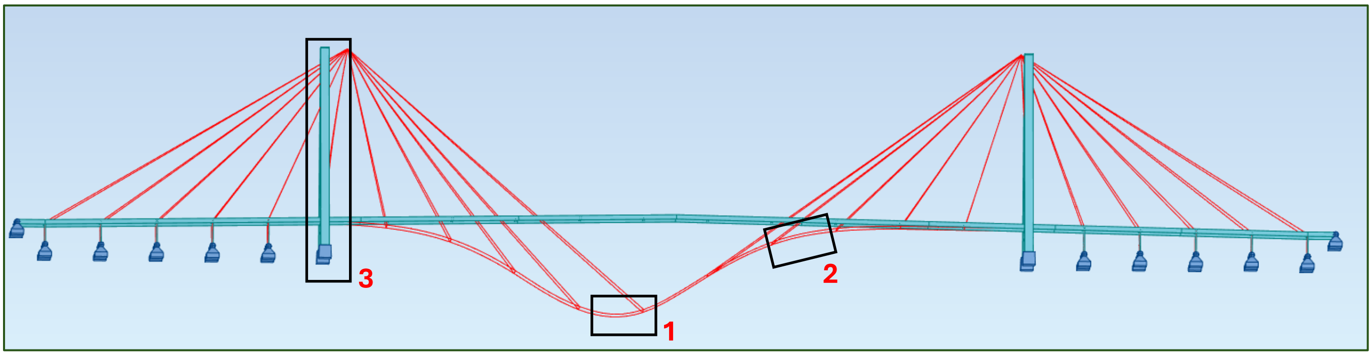
| Member/Node/Case | FX (kN) |
|---|---|
| 74/ 33/ 1 | -78.48 |



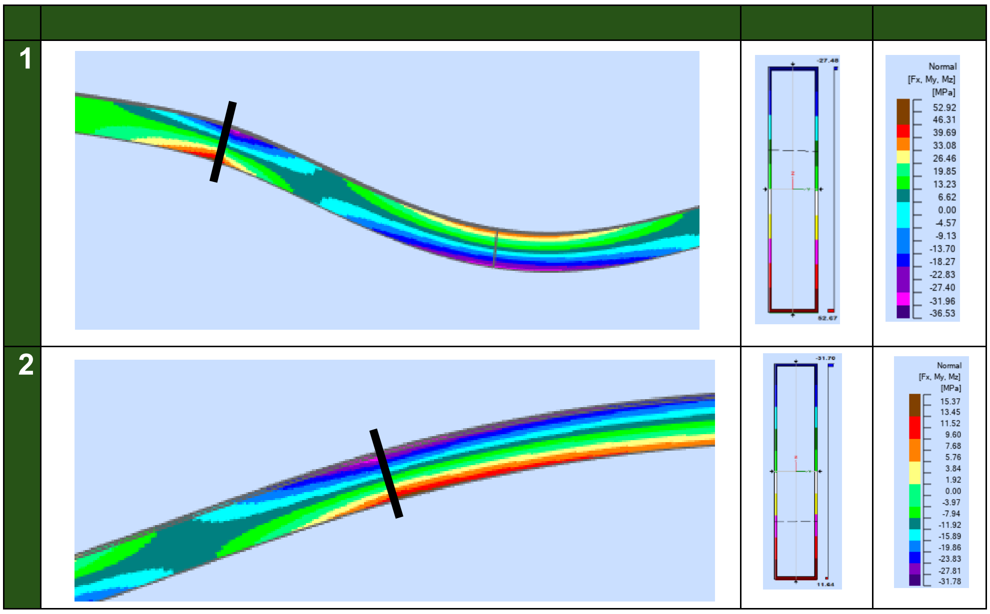

Edge Beam Bending Moment Distribution

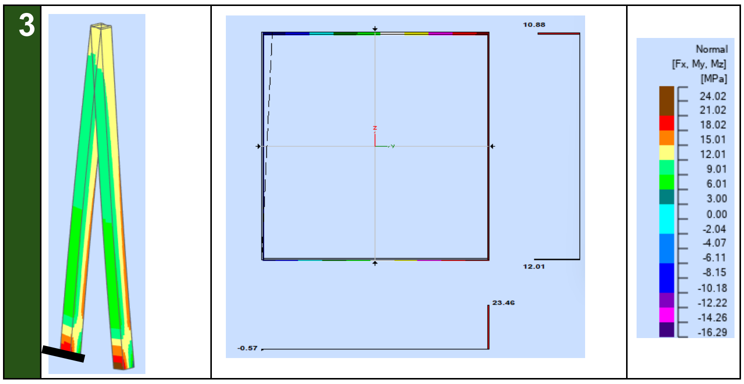
## 4.2 Asymmetric Concentrated Crowd Loading

### 4.2.1 Deflection Based Location of Interest Identification




### 4.2.2 Maximum Stress Locations



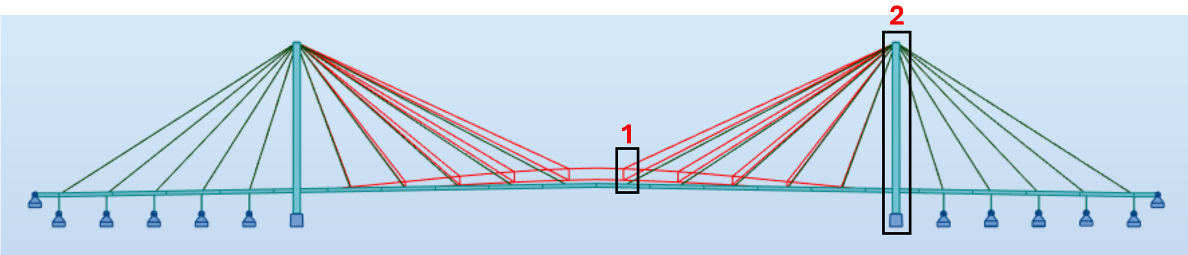

Our initial calculations map well where the maximum stress locations will occur. The FEA gives us maximum stress in the edge beams of 52.9MPa which is close to that of our calculated predictions of 62.9MPa. The differences could either be due to an overprediction from our initial analysis, for example not considering the plastic redistribution of bending stresses along the edge beams. Or it could be due to a lack of mesh density at the critical locations. However, we progressively increased this density to ascertain if the value would converge with our prediction. This resulted in the 52.9MPa value with no movement upwards towards the prediction.

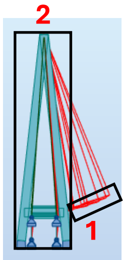
The 24.02Mpa maximum stress value produced by the FEA in the antenna is higher than our expectation of 16.7MPa. We think this is likely due to the localised stress concentrations at the fixed connections and interface with the deck.

## 4.3 Winter Storm: Combined Wind & Snow Loading

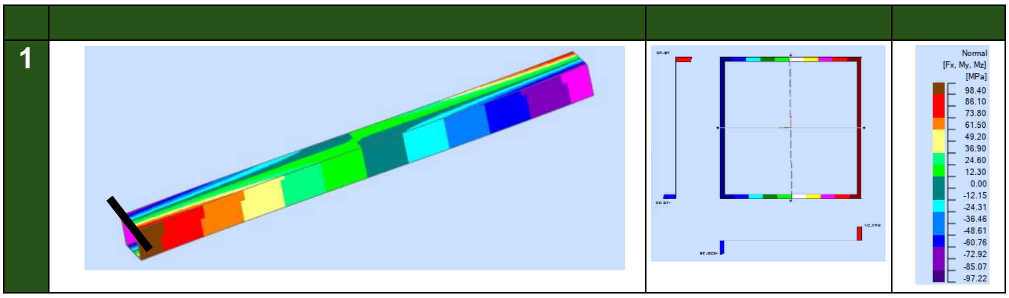
### 4.3.1 Deflection Based Location of Interest Identification

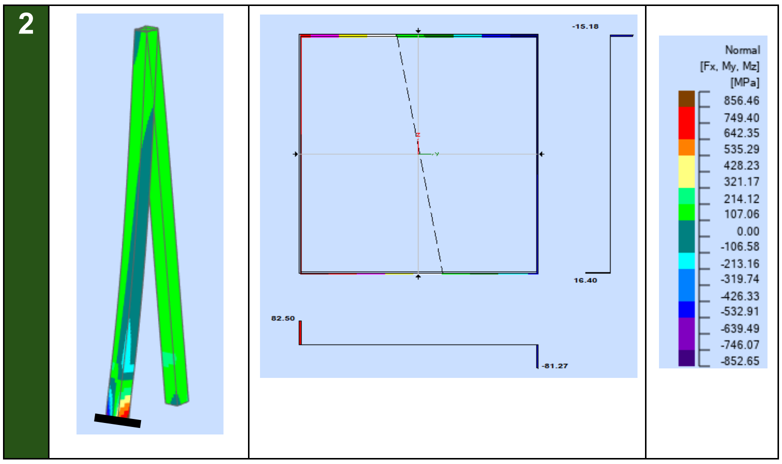
Appropriate snow and wind loads were calculated based on ***Eurocode 1991-1 (Actions on Structures)***. The snow load was calculated as 0.681kN/m^2 (1.02kN/m^2 factored). The wind load was calculated as 0.781kN/m^2 (1.17kN/m^2 factored). These were both applied to the structure but were not combined with the usual live loading as it is unlikely that there would be people on the bridge in a severe storm event. The deflected shapes produced are shown on the next page.




### 4.3.2 Maximum Stress Locations




In this model, the maximum stress occurs in the primary beams connecting the two edge beams with a value of 98.4MPa, In our initial calculations we did not account for the primary beams acting as supports for the edge beams and therefore our maximum stress was predicted to be exceptionally higher at a value of 6.3x10^3KPa due to the gross oversimplification of taking the whole deck as unsupported in the Y plane between the antennae. The maximum stress in the antenna of 856KPa highlights the susceptibility of the FEA to local extreme maximum stress values. Consequently, the rest of the plot's contouring is skewed towards this value. This means that other local maxima and minima are not shown correctly as the gaps in the contours are too great.  Additionally, it is also not very close to our calculated value of 46.3KPa as the effect of the primary beams linking the two edge beams together leads to severe torsional buckling effects on the windward antenna.

## 5.0 Summary & Conclusions

We set ourselves a big challenge as a team to model such a complicated structure using finite element software. We underestimated the complexity of modelling the geometry accurately in 3D however we still hold the belief that it was necessary due to the form of this bridge. Additionally, the high level of static indeterminacy present in cable stay bridges such as this one posed challenges for us when we were attempting to verify the results produced by the FEA using our hand calculations.

The overall structural behavior of the bridge was replicated well, with the deflected shapes, stress and bending moment distributions being in line with our expectations. However, we struggled to verify some of the results from the model, especially on the antennae. Our hand calculations were unable to predict the localized stress concentrations such as at the cable-deck connection nodes and the fixed bases of the antennae. Overall, we are happy with the fact that despite the complexities our FEA managed to accurately model the stresses in critical locations around the structure.

Advanced Polymer Composites for Structural Applications in Construction, n.d. The design, construction and in-service performance of the all-composite Aberfeldy footbridge [Online]. p.pg 445-453. Available from: https://doi.org/10.1680/apcfsaic.31227 [Accessed 24 May 2021].

Quadrino, A., Damiani, M., Penna, R., Feo, L. and Nisticò, N., 2021. Design of an FRP Cable-Stayed Pedestrian Bridge. Morphology, Technology and Required Performances [Online]. *Lecture notes in civil engineering*, pp.46–62. Available from: https://doi.org/10.1007/978-3-030-88166-5_4 [Accessed 12 December 2024].